WEB SCRAPING WITH PYTHON

UNIT-1

Building Scrapers: First Web Scraper: Connecting, An Introduction to BeautifulSoup, Advanced HTML Parsing: Another Serving of BeautifulSoup, Regular Expressions, Regular Expressions and BeautifulSoup, Accessing Attributes, Lambda Expressions, Writing Web Crawlers: Traversing a Single Domain, Crawling an Entire Site, Crawling Acrossthe Internet.

UNIT-II

Web Crawling Models: Planning and Defining Objects, Dealing with Different Website Layouts, Structuring Crawlers. Scrapy: Installing Scrapy, Initializing a New Spider, Writing a Simple Scraper, Spidering with Rules, Creating Items, Outputting Items, The Item Pipeline, Logging with Scrapy. Storing Data: Media Files, Storing Data to CSV, MySQL: Integrating with Python, Database Techniques and Good Practice, Six Degrees in MySQL, Email

UNIT-III

Advanced Scraping:Reading Documents: Document Encoding, Text, CSV, PDF, Microsoft Word and .docx,Reading and Writing Natural Languages: Summarizing Data, Markov Models, Natural Language Toolkit. Crawling Through Forms and Logins: Python Requests Library, Submitting a Basic Form, Radio Buttons, Checkboxes, and Other Inputs, Submitting Files and Images, Handling Logins and Cookies.

UNIT-IV

Crawling Through APIs: A Brief Introduction to APIs, Parsing JSON, Undocumented APIs, Finding Undocumented APIs, Documenting Undocumented APIs, Finding and Documenting APIs Automatically.

Image Processing and Text Recognition: Overview of Libraries, Pillow, Processing Well-Formatted Text, Reading CAPTCHAs and Training Tesseract, Retrieving CAPTCHAs and Submitting Solutions

UNIT-V

Avoiding Scraping Traps: A Note on Ethics, Looking Like a Human, Common Form Security Features, The Human Checklist Testing Your Website with Scrapers: An Introduction to Testing, Python unit test, Testing with Selenium, unittest or Selenium. The Legalities and Ethics of Web Scraping: Trademarks, Copyrights, Patents, Trespass to Chattels, The ComputerFraud and Abuse Act, robots.txt and Terms of Service, Three Web ScraperS